bxcjkjatype_bxcjkvert_4762decd | compact_paper | 2 page(s)

Package: bxcjkjatype
Size: medium, 119 nonblank lines, 4790 chars
Tags: color, crossrefs, custom_environments, custom_macros, fonts_symbols, footnotes, inline_math, lists, text_sectioning
Source: /usr/local/texlive/2026/texmf-dist/doc/latex/bxcjkjatype/bxcjkvert.tex

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The bxcjkvert Package

Takayuki YATO (aka. "ZR")

v0.5 [2023/07/23]

Abstract

This package is a tailored version of the CJKvert package, accommodated for Japanese typesetting, where mixture of horizontal and vertical writings is common.

Contents

1 Package Loading

`\usepackage[<option>,...]{bxcjkvert}`

The CJKvert package will be automatically loaded, if not yet loaded.
The available options are described hereafter.

1.1 Options for configuring font reset

When a command for switching direction (`\CJKhorz` or `\CJKvert`) is invoked, CJKvert resets the current font parameter by issuing `\normalfont`. This feature is inconvenient, particularly when authors mix different writing directions. Thus bxcjkvert suppresses the feature by default, but it can be configured by the `resetfont` option.

- `resetfont=true`: Makes direction comamnds reset the current font by issuing the `\normalfont` command.

  Note: This is the original behavior of CJKvert.

- `resetfont=false` (default): Makes direction comamnds retain the current font.

1.2 Options for configuring adjustment of the baseline stretch

CJKvert makes some adjustment to the value of `\baselinestretch` when the writing direction is changed. Namely it makes the baseline stretch enlarged by the factor of `\CJKbaselinestretch`[1] when `\CJKvert` is used.

[1]The value of `\CJKbaselinestretch` is 1.3 by default.

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However as far as Japanese typesetting is concerned, there is no need to tweak the baseline stretch value. Thus bxcjkvert suppresses this feature, but again it can be configured by the `usebaselinestretch` option.

- `usebaselinestretch=true`: This is the same as the behavior of CJKvert *with* the `usebaselinestretch` option. When `\CJKvert` is invoked, the baseline stretch will be multiplied by the value of `\CJKbaselinestretch`. When `\CJKhorz` is invoked, the baseline stretch will be reverted.[2]

- `usebaselinestretch=false`: This is the same as the behavior of CJKvert *without* the `usebaselinestretch` option. When `\CJKvert` is invoked, the baseline stretch will be set to the value of `\CJKbaselinestretch`. When `\CJKhorz` is invoked, the baseline stretch will be reset to 1. This option ignores the user's setting to the baseline stretch.

- `usebaselinestretch=retain` (default): Makes direction comamnds leave the baseline stretch unchanged.

Note: If CJKvert is loaded with the `usebaselinestretch` option in advance, then the value of `usebaselinestretch` of this package will default to `true` instead of `retain`.

1.3 Options for configuring initial writing direction

CJKvert sets the initial writing direction of the document to vertical. But bxcjkvert allows authors to choose the initial direction.

- `main=true`: Sets the initial direction to vertical.

  Note: This is the original behavior of CJKvert.

- `main=false`: Sets the initial direction to horizontal.

- `main=retain` (default): Does not specify the initial direction. In this case, authors can use `\CJKvert` or `\CJKhorz` in the preamble to decide the initial direction.

[2]Note that, however, in this case the baseline stretch will be reverted to the value at the time when CJKvert was loaded. it might be against authors' expectation.

correctmathalign_correctmathalign_f9cff2e2 | math_short | 2 page(s)

Package: correctmathalign
Size: medium, 99 nonblank lines, 3079 chars
Tags: aligned_math, captions, crossrefs, custom_macros, display_math, figures_images, lists, minipage_boxes, text_sectioning
Source: /usr/local/texlive/2026/texmf-dist/doc/latex/correctmathalign/correctmathalign.tex

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Correction Spacings: eqnarray(*), aligned, alignedat, and gathered

Yuwsuke Kieda

2017/04/04 v1.1

1 Descriptions

We will realignment some expression environments spacing. It is a correction in the horizontal spacing of the alignment.

2 Usage

2.1 Preamble

```
\usepackage{amsmath}
\usepackage{correctmathalign}
```

Remark: amsmath package is optional: if you use aligned, alignedat, or gathered environments.

Remark: The problem with the amsmath package (2016/11/05 2.16a) was solved by the original. See the options `alignedleftspaceyes`, `alignedleftspaceno`, and `alignedleftspaceyesifneg`. We dealt with versions prior to 2016/03/10 v2.15b.

2.2 Options

- `latexorg`: original behavior of LaTeX/amsmath package
- `fleqn`: corresponds to class file and/or amsmath package's fleqn option

3 File with Original Definitions

- `eqnarray`: latex.ltx
- `\start@aligned` and `\gathered`: amsmath.sty (before 2016/03/10 v2.15b)

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$$\mathrm{I} = \mathrm{H}(x+1)$$

$$\begin{array}{rcl} \mathrm{I} & = & \mathrm{H}(x) \\ & = & \mathrm{H}(x+1) \end{array}$$

Figure 1: `eqnarray*` environment by LaTeX default.

$$\mathrm{I} \begin{aligned}[t] &= \mathrm{H}(x) \\ &= \mathrm{H}(x+1) \end{aligned}$$

$$\begin{aligned} \mathrm{I} &= \mathrm{H}(x) \\ &= \mathrm{H}(x+1) \end{aligned}$$

Figure 2: `aligned` environment by `amsmath.sty` default (before 2016/03/10 v2.15b).

4 License

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5 Samples

Cf. defaults of LaTeX and `amsmath.sty` before 2016/03/10 v2.15b (Figures 1 and 2).

$$\mathrm{I} = \mathrm{H}(x+1)$$

$$\begin{aligned} \mathrm{I} &= \mathrm{H}(x) \\ &= \mathrm{H}(x+1) \end{aligned}$$

$$\mathrm{I} \begin{aligned}[t] &= \mathrm{H}(x) \\ &= \mathrm{H}(x+1) \end{aligned}$$

$$\begin{aligned} \mathrm{I} &= \mathrm{H}(x) \\ &= \mathrm{H}(x+1) \end{aligned}$$

5.1 Use LaTeX code

```
\[
 \mathrm{I} = \mathrm{H}(x + 1)
\]
\begin{eqnarray*}
 \mathrm{I} &=& \mathrm{H}(x)\\
           &=& \mathrm{H}(x + 1)
\end{eqnarray*}

\begin{align*}
 \mathrm{I} & \begin{aligned}[t]
               &= \mathrm{H}(x)\\
               &= \mathrm{H}(x + 1)
              \end{aligned}\\
 \mathrm{I} &= \mathrm{H}(x)\\
           &= \mathrm{H}(x + 1)
\end{align*}
```

minutes_overview_893facd9 | package_example | 2 page(s)

Package: minutes
Size: medium, 146 nonblank lines, 6518 chars
Tags: complex_table, crossrefs, custom_macros, diagram_nodes_edges, inline_math, lists, long_table, multi_column, text_sectioning
Source: /usr/local/texlive/2026/texmf-dist/doc/latex/minutes/Overview.tex

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Options

Option	description	Beschreibung
ListTitle	List-like header	Listartiger Titel
TableTitle	Table-like header	Tabellenartiger Titel
OneColumn	One column output	Einspaltige Ausgabe
TwoColumn	Two column output	Zweispaltige Ausgabe
CreateCld	create cld-File	Erzeugung einer cld-Datei
8+3	filenames 8+3	8+3 Dateinamen
Fileinfo	information in lists	Ausgabe von Dateiinformationen
Secret	Print secret parts	Ausgabe geheimer Teile
ASCII	Prepared for ASCII	Ausgabe für reines ASCII
Other	Option from `minutes.cfg`	Option aus `minutes.cfg`

Commands

english command	deutsches Kommando	Parameter
Minutes	**Protokoll**	
`\begin{Minutes}`	`\begin{Protokoll}`	`{Title/Titel}`
`\maketitle`	`\protokollKopf`	
`\foreignMinutes`	`\fremdProtokoll`	
`\end{Minutes}`	`\end{Protokoll}`	
Header data	**Kopfdaten**	
`\subtitle`	`\untertitel`	`{text}`
`\moderation`	`\moderation`	`{Name}`
`\minutetaker`	`\protokollant`	`{Name}`
`\participant`	`\teilnehmer`	`{Name,Name}`
`\guest`	`\gaeste`	`{Name,Name}`
`\minutesdate`	`\sitzungsdatum`	`{Date/Datum}`
`\starttime`	`\sitzungsbeginn`	`{time/Zeit}`
`\endtime`	`\sitzungsende`	`{time/Zeit}`
`\location`	`\sitzungsort`	`{location/Ort}`
`\cc`	`\verteiler`	`{Name,Name}`
`\missing`	`\fehlend`	`[Name]{Name,Name}`
`\missingExcused`	`\fehlendEntschuldigt`	`{Name,Name}`
`\missingNoExcuse`	`\fehlendUnentschuldigt`	`{Name,Name}`
sectioning	**Gliederung**	
`\topic`	`\topic`	`[short]{Title}`
`\addtopic`	`\zusatztopic`	`[short]{Title}`
`\subtopic`	`\subtopic`	`[short]{Title}`
[]=optional	{}=must be filled/Mussfeld	{*}= {} or */{} oder *

p2

english command	deutsches Kommando	Parameter
`\subsubtopic`	`\subsubtopic`	`[short]{Title}`
`\minitopic`	`\minitopic`	`{Title}`
`\newcols`	`\neueSpalte`	`[Title][1]`
Tasks	**Aufgaben**	
`\task`	`\aufgabe`	`[done]{*who}[when]{what}`
`\listoftasks`	`\aufgabenliste`	`[file]`
Dates	**Termine**	
`\schedule`	`\termin`	`[cal]{when}[time]{what}[long]`
Voting	**Abstimmungen**	
`\begin{Vote}`	`\begin{Abstimmung}`	
`\vote`	`\abstimmung`	`{text}{yes}{no}{-}[text]`
`\end{Vote}`	`\end{Abstimmung}`	
Decisions	**Entscheidungen**	
`\decisionTheme`	`\beschlussThema`	`{ref}{Title}`
`\decision`	`\beschluss`	`{*ref}{Text}[longText]`
`\listofdecisions`	`\beschlussliste`	
Argumentations	**Argumentationen**	
`\begin{Argumentation}`	`\begin{Argumentation}`	
`\pro`	`\pro`	
`\Pro`	`\Pro`	
`\contra`	`\contra`	
`\Contra`	`\Contra`	
`\result`	`\ergebnis`	
`\end{Argumentation}`	`\end{Argumentation}`	
`\begin{Opinions}`	`\begin{Meinungen}`	
`\end{Opinions}`	`\end{Meinungen}`	
`\opinion`	`\meinung`	`{who}{what}`
Versions	**Versionen**	
`\begin{Secret}`	`\begin{Geheim}`	
`\end{Secret}`	`\end{Geheim}`	
`\secret`	`\geheim`	`{text}`
Attachments	**Anhänge**	
`\attachment`	`\anhang`	`[ref]{Title}{pages}`
`\listofattachments`	`\anhangsliste`	
`\postscript`	`\nachtrag`	`{text}`
Other commands	**Sonstige Kommandos**	
`\inputminutes`		`{file}`
[]=optional	{}=must be filled/Mussfeld	{*}= {} or */{} oder *

sciposter_sciposter_example_e1a634cd | poster_layout | 1 page(s)

Package: sciposter

Size: large, 280 nonblank lines, 12749 chars

Tags: algorithm_pseudocode, aligned_math, captions, citations_bibliography, color, crossrefs, custom_macros, display_math, figures_images, fonts_symbols, inline_math, lists, multi_column, poster_blocks, simple_table, text_sectioning, theorems_proofs

Source: /usr/local/texlive/2026/texmf-dist/doc/latex/sciposter/sciposterexample/sciposter-example.tex

p1

Generalized Pattern Spectra Sensitive to Spatial Information

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Abstract

Morphological pattern spectra computed from granulometries are frequently used to classify the size classes of details in textures and images. An extension of this technique, which retains information on the spatial distribution of the details in each size class is developed. Algorithms for computation of these spatial pattern spectra for a large number of granulometries on binary images are presented.

1. Introduction

GRANULOMETRIES are ordered sets of morphological openings or closings, each of which removes image details below a certain size. These can be used for texture analysis through the use of *pattern spectra*, which show how the number of foreground pixels in the image changes as a function of the size parameter [3]. A drawback of the classical definition of pattern spectra is that spatial information is not included in a pattern spectrum as shown below. In this paper, *spatial pattern spectra* are developed which retain information on the distribution of these details at different scales.

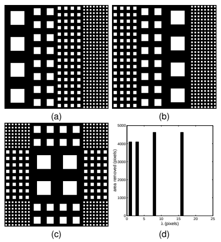

Figure 1: *Parts (a) through (c) show three images consisting of squares of different sizes; (d) shows the pattern spectra, denoting the number of foreground pixels removed by openings by reconstruction by $\lambda \times \lambda$ squares. No granulometry is capable of separating the patterns, because the only differences between the images lie in the distributions of the connected components.*

2. Theory

Let binary images X and Y be defined as a subset of the image domain $\mathbf{M} \subset \mathbb{Z}^n$ or $\mathbb{R}^n$ (usually $n = 2$).

Definition 1 *A binary granulometry is a set of operators $\{\alpha_r\}$ with r from some ordered set Λ (usually $\Lambda \subset \mathbb{R}$ or $\mathbb{Z}$), with the following three properties*

$$\alpha_r(X) \subset X \tag{1}$$
$$X \subset Y \Rightarrow \alpha_r(X) \subset \alpha_r(Y) \tag{2}$$
$$\alpha_r(\alpha_s(X)) = \alpha_{\max(r,s)}(X), \tag{3}$$

for all $r, s \in \Lambda$.

Definition 2 *The pattern spectrum $s_\alpha(X)$ obtained by applying granulometry $\{\alpha_r\}$ to a binary image X is defined as*

$$(s_\alpha(X))(u) = -\left.\frac{\partial A(\alpha_r(X))}{\partial r}\right|_{r=u} \tag{4}$$

in which $A(X)$ is a function denoting the Lebesgue measure in $\mathbb{R}^n$.

In the case of discrete images, and with $r \in \Lambda \subset \mathbb{Z}$, this differentiation reduces to

$$\begin{aligned}(s_\alpha(X))(r) &= \#(\alpha_r(X) \setminus \alpha_{r^+}(X)) &(5)\\ &= \#(\alpha_r(X)) - \#(\alpha_{r^+}(X)), &(6)\end{aligned}$$

with $r^+ = \min\{r' \in \Lambda | r' > r\}$, and $\#(X)$ the numnber of elements of X.

The opening transform [5] Ω_X of a binary image X for a granulometry α_r is

$$\Omega_X(x) = \max\{r \in \Lambda | x \in \alpha_r(X)\} \tag{7}$$

The pattern spectrum of a binary image X using granulometry $\{\alpha_r\}$ is the histogram of Ω_X obtained with the same size distribution [5], disregarding the bin for grey level 0.

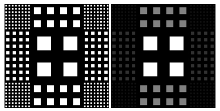

Figure 2: *Opening transform with $\{\alpha_r\}$ as in Fig. 1: (left) original image; (right) opening transform (contrast stretched for clarity).*

3. Spatial pattern spectra

Pattern spectra only retain the amount of detail present at scale r. This can be amended by computing some parameterization of the spatial distribution in an image $\alpha_r(X) \setminus \alpha_{r+}(X)$ as a function of r.

Definition 3 *Let $M(X)$ be some parameterization of the spatial distribution of detail in the image X. The spatial pattern spectrum $S_{M,\alpha}$ is then defined as*

$$(S_{M,\alpha}(X))(r) = M(\alpha_r(X) \setminus \alpha_{r+}(X)). \tag{8}$$

An obvious parameterization of the spatial distribution is through the use of moments. Focusing on the case of 2-D binary images, the moment m_{ij} of order ij of an image X is given by

$$m_{ij}(X) = \sum_{(x,y)\in\mathbf{X}} x^i y^j. \tag{9}$$

The spatial moment spectrum $S_{m_{ij},\alpha}$ of order ij is

$$(S_{m_{ij},\alpha}(X))(r) = m_{i,j}(\alpha_r(X) \setminus \alpha_{r^+}(X)). \tag{10}$$

For $i = 0$ and $j = 0$ we obtain the standard pattern spectrum. For each r, $(S_{m_{ij},\alpha}(X))(r)$ is just the moment of an image, therefore, derived parameters such as coordinates of the centre of mass, (co-)variances, skewness and kurtosis of the distribution of details at each scale can be computed easily. We can then define pattern mean spectra, pattern (co-)variance spectra, pattern kurtosis spectra, etc. The pattern mean-x and variance-x spectra ($S_{\bar{x},\alpha}$ and $S_{\sigma(x),\alpha}$) are defined as:

$$S_{\bar{x},\alpha} = \frac{S_{m_{10},\alpha}}{S_{m_{00},\alpha}} \tag{11}$$

and

$$S_{\sigma(x),\alpha} = \sqrt{\frac{S_{m_{20},\alpha}}{S_{m_{00},\alpha}} - S_{\bar{x},\alpha}}. \tag{12}$$

These two are shown in Figures 3 and 4. Note that these definitions hold only where $(S_{m_{00},\alpha}(f))(r) \neq 0$. For all other values of r they will be defined as zero. Further post-processing can be done to compute central moments and moment invariant from pattern moment spectra [1, 2].

4. An Algorithm

Nacken [5] derived an algorithm for computation of pattern spectra for granulometries based on openings by discs of increasing radius for various metrics, using the opening transform. After the opening transform has been computed, it is straightforward to compute the pattern spectrum:

- Set all elements of array $\mathtt{S}$ to zero
- For all $x \in X$ increment $\mathtt{S}[\Omega_X(x)]$ by one.

To compute the pattern *moment* spectrum, the only thing that needs to be changed is the way $\mathtt{S}[\Omega_X(x)]$ is incremented. As shown in Algorithm 1.

- Set all elements of array $\mathtt{S}$ to zero
- For all $(x, y) \in X$ increment $\mathtt{S}[\Omega_X(x, y)]$ by $x^i y^j$.

Algorithm 1: *Algorithm for computation of pattern moment spectrum of order ij.*

This algorithm can readily be adapted to other granulometries, simply by computing the appropriate opening transform.

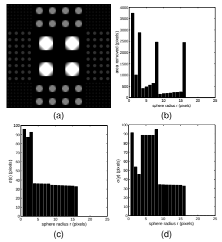

Figure 3: *The opening transform using city-block metric: (a) opening transform of Fig. 1(c); (b) pattern spectrum; (c) pattern variance-x; (d) variance-y spectra.*

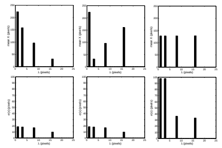

Figure 4: *Pattern mean-x (top) and variance-x (bottom) spectra: the three collumns show spectra for Fig. 1(a), (b) and (c) from left to right respectively. Unlike the standard pattern spectra, these spatial pattern spectra can distinguish the three images.*

5. Discussion

Spatial pattern spectra form a useful supplement to ordinary pattern spectra, because of their ability to retain spatial information. Pattern moment spectra, in particular, are easily computed concurrently with computation of the standard pattern spectrum. Post-processing of these pattern moment spectra can be done to yield a number of easily interpreted spectra, such as pattern mean, variance, skew, and kurtosis spectra, which have reduced covariance compared to the "raw" pattern moment spectra. Invariance to rotation, translation or scale change can also be achieved by post-processing [1, 2].

In the future grey scale versions of these spatial pattern spectra will be developed. I expect that the efficient grey level algorithms for area and attribute pattern spectra [4] can be adapted to spatial pattern spectra as well.

References

[1] J. Flusser and T. Suk. Pattern recognition by affine moment invariants. *Pattern Recognition*, 26:167–174, 1993.

[2] M. K. Hu. Visual pattern recognition by moment invariants. *IRE Transactions on Information Theory*, IT-8:179–187, 1962.

[3] P. Maragos. Pattern spectrum and multiscale shape representation. *IEEE Trans. Patt. Anal. Mach. Intell.*, 11:701–715, 1989.

[4] A. Meijster and M. H. F. Wilkinson. A comparison of algorithms for connected set openings and closings. *IEEE Trans. Patt. Anal. Mach. Intell.*, 24(4):484–494, 2002.

[5] P. F. M. Nacken. *Image Analysis Methods Based on Hierarchies of Graphs and Multi-Scale Mathematical Morphology*. PhD thesis, University of Amsterdam, Amsterdam, The Netherlands, 1994.

ICPR 2002, 16th International Conference on Pattern Recognition, 11-15 August 2002, Québec City, Canada